

Technology Stack

Date	1 October 2026
Team ID	SWTID-2026-7957
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

EduGenie is a lightweight, AI-powered educational assistant that helps students and self-learners understand difficult topics. It answers academic questions, explains concepts in simple words, generates three-question quizzes, summarizes long passages and builds step-by-step learning paths from beginner to advanced level. The backend is an asynchronous FastAPI application and the frontend is a simple HTML, CSS and JavaScript page served with Jinja2, so it runs smoothly even on ordinary laptops.

To balance speed, cost and quality, EduGenie uses a hybrid AI approach: the Google Gemini API provides cloud intelligence, while the small LaMini-Flan-T5 model runs locally for concept explanations. The code is managed with Git and GitHub and deployed on Vercel. The application does not use a database because it stores no user data.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML, CSS, JavaScript, Jinja2 Templates	A lightweight, responsive single page that loads instantly with no heavy framework. Jinja2 serves the page from the backend, and plain JavaScript calls the API and shows results, including quiz feedback.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3.10+, FastAPI, Uvicorn	Fast and asynchronous, with minimal boilerplate and automatic API documentation at /docs. It suits many concurrent AI requests, and Python gives direct access to the AI libraries.
3	AI / ML Layer	Google Gemini (Google GenAI SDK); LaMini-Flan-T5-783M with Hugging Face Transformers and PyTorch	Gemini handles question answering, quizzes, summaries and learning paths in the cloud. The small, instruction-tuned LaMini model explains concepts locally on an ordinary CPU, and Gemini is the fallback if it cannot load.
4	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	None (stateless application)	The first release stores no user data, which keeps the app simple and private. Quiz answers are held in the browser while the page is open. A database such as SQLite or PostgreSQL can be added later for progress tracking.
5	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Vercel (live deployment); local machine with Uvicorn (development)	Vercel gives free, quick deployment of the web app straight from GitHub. The API key is stored in Vercel's environment settings. Locally the app runs with one command, <code>uvicorn main:app --reload</code> .
6	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git and GitHub	Lets the five team members collaborate, keeps a full history of changes and triggers deployments.

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7	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Google AI Studio (Gemini API key), python-dotenv, Visual Studio Code	Google AI Studio issues the Gemini API key. python-dotenv loads the key from a .env file, and VS Code with a Python virtual environment is the development setup.